MICHIGAN TECH · SPRING 2027

Wanted: Future Autonomy Stack Engineers.



EE5531 Introduction to Robotics

Transform **raw sensor streams** into **autonomous movement**. You will build software to map environments, estimate position under uncertainty, and navigate physical platforms using algorithmic foundations standard in industry.



Built by students: 3D LiDAR map of Husky Plaza, about 2 million points

Real robots

Clearpath Jackals and TurtleBot3s with LiDAR and IMUs. Not just simulation.

The industry toolchain

ROS 2, Nav2, SLAM, Gazebo, Python, and Git. If you've read an AV job posting, you've seen this list.

The math behind autonomy

Derive a Kalman filter, then watch it pull a drifting robot back on track.

A portfolio you can show

Public GitHub repos with code, maps, and video. Exactly what today's employers expect.

THE COURSE FOLLOWS THE LOOP EVERY AUTONOMOUS SYSTEM RUNS

Observe — Fit a beam model to your own LiDAR

› **Orient** — Kalman filters and SLAM maps of real rooms

› **Decide** — Nav2 planners, costmaps, and recovery

› **Act** — Your full stack on a Jackal, scored

— WHERE EE5531 ALUMNI WORK NOW —

May mobility · TORC · gm general motors · TORO · Qualcomm

...and throughout the autonomous vehicle sector.

LECTURE	Mon & Wed, 1:00–1:50 PM
PROJECTS	Mon, 2:00–3:50 PM
CRN	12485 / 12486
PREREQS	FOSS101 or equivalent experience
FOR	MTU Robotics Engineering BS grads, and anyone with Linux, Python/C++, and Git experience.
CONTACT	Dr. Jeremy Bos · jpbos@mtu.edu

Runs only with enough enrollment. Register early and bring a friend.

Scan for the full course site

robust-autonomous-systems-laboratory.github.io/ee5531